

Advanced (Habanero)

- Q1.** What is the effect of the transformation $y = f(x) + 3$ on the graph of $y = f(x)$?
(A) Shift down 3 units
(B) Shift up 3 units
(C) Shift left 3 units
(D) Shift right 3 units
(E) No change
- Q2.** Which transformation reflects the graph of $y = f(x)$ over the x-axis?
(A) $y = f(x) + 2$
(B) $y = f(x) - 2$
(C) $y = -f(x)$
(D) $y = f(-x)$
(E) $y = 2f(x)$
- Q3.** What is the effect of the transformation $y = f(2x)$ on the graph of $y = f(x)$?
(A) Horizontal stretch by a factor of 2
(B) Horizontal compression by a factor of 2
(C) Vertical stretch by a factor of 2
(D) Vertical compression by a factor of 2
(E) No change
- Q4.** Which transformation represents a vertical stretch of the graph of $y = f(x)$ by a factor of 3?
(A) $y = 3f(x)$
(B) $y = f(3x)$
(C) $y = f(x) + 3$
(D) $y = f(x) - 3$
(E) $y = -f(x)$
- Q5.** What is the effect of the transformation $y = f(x - 2)$ on the graph of $y = f(x)$?
(A) Shift left 2 units
(B) Shift right 2 units
(C) Shift up 2 units
(D) Shift down 2 units
(E) No change
- Q6.** Which transformation represents a reflection of the graph of $y = f(x)$ over the y-axis?
(A) $y = f(-x)$
(B) $y = -f(x)$
(C) $y = f(x) + 2$
(D) $y = f(x) - 2$
(E) $y = 2f(x)$
- Q7.** What is the effect of the transformation $y = -2f(x)$ on the graph of $y = f(x)$?
(A) Vertical stretch by a factor of 2 and reflection over x-axis
(B) Vertical compression by a factor of 2 and reflection over x-axis
(C) Horizontal stretch by a factor of 2
(D) Horizontal compression by a factor of 2
(E) No change
- Q8.** Which transformation represents a horizontal stretch of the graph of $y = f(x)$ by a factor of 2?
(A) $y = f(2x)$
(B) $y = f(x/2)$
(C) $y = 2f(x)$
(D) $y = f(x) + 2$
(E) $y = f(x) - 2$

Open Ended Questions (FRQ)

- Q9.** Graph the function $y = 2f(x - 1) - 3$ if $f(x) = x^2$. Label the vertex and any x-intercepts.
- Q10.** Compare the graphs of $y = f(x)$, $y = f(x + 2)$, and $y = f(x + 2) - 1$. Describe the transformations that occur between each pair of graphs.

Chef's Tips

- Remind students to show all work and explain their reasoning for FRQs.
- Allow students to use graphing calculators to check their work.
- Encourage students to use the parent function to help with transformations.